

'NumPy' : 'Numerical Python'

NumPy is a Python's library which is use to perform different mathematical operations on arrays

Array is denoted with '['].It is ordered (index position is fixed), mutable (items can be modified) and homogenous (items of similar datatypes) collection of items

```
In [1]: import numpy as np
```

```
In [3]: #creating an 1-D array
a=np.array([1,2,3,4,5])
a
```

```
Out[3]: array([1, 2, 3, 4, 5])
```

```
In [15]: #tuple to array
t=np.array((1,2,3,4))
t
```

```
Out[15]: array([1, 2, 3, 4])
```

```
In [7]: #set to array
x=np.array({1,2,3,4})
x
```

```
Out[7]: array({1, 2, 3, 4}, dtype=object)
```

```
In [9]: #dictionary to array
d=np.array({'a':1,'b':2,'c':3,'d':4})
d
```

```
Out[9]: array({'a': 1, 'b': 2, 'c': 3, 'd': 4}, dtype=object)
```

```
In [11]: #creating a 2-D array
a=np.array([[1,3,5],[2,4,6]])
a
```

```
Out[11]: array([[1, 3, 5],
               [2, 4, 6]])
```

```
In [19]: t=np.array([(1,2,3,4),(8,9,7,6)])
t
```

```
Out[19]: array([[1, 2, 3, 4],
               [8, 9, 7, 6]])
```

```
In [21]: b=np.array([(1,2,3,4),(8,9,7,6)])
b
```

```
Out[21]: array([[1, 2, 3, 4],
               [8, 9, 7, 6]])
```

```
In [23]: b=np.array([[1,2,3,4],[5,6,7,8],[9,3,6,4]])  
b
```

```
Out[23]: array([[1, 2, 3, 4],  
               [5, 6, 7, 8],  
               [9, 3, 6, 4]])
```

```
In [25]: #Array attributes  
#shape  
b.shape
```

```
Out[25]: (3, 4)
```

```
In [27]: #size  
b.size
```

```
Out[27]: 12
```

```
In [29]: #ndim  
b.ndim
```

```
Out[29]: 2
```

```
In [31]: #import numpy module  
import numpy as np
```

```
In [33]: #arange  
np.arange(1,12)
```

```
Out[33]: array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11])
```

```
In [35]: #zeros()  
np.zeros(5)
```

```
Out[35]: array([0., 0., 0., 0., 0.])
```

```
In [39]: np.zeros((4,5))
```

```
Out[39]: array([[0., 0., 0., 0., 0.],  
               [0., 0., 0., 0., 0.],  
               [0., 0., 0., 0., 0.],  
               [0., 0., 0., 0., 0.]])
```

```
In [49]: np.zeros((4,5)).astype(int)
```

```
Out[49]: array([[0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0]])
```

```
In [51]: np.zeros((4,5),dtype='int')
```

```
Out[51]: array([[0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0]])
```

```
In [53]: np.zeros((4,5),dtype=int)
```

```
Out[53]: array([[0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0],
                [0, 0, 0, 0, 0]])
```

```
In [55]: #ones
np.ones(5)
```

```
Out[55]: array([1., 1., 1., 1., 1.])
```

```
In [57]: np.ones((4,5))
```

```
Out[57]: array([[1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1.],
                [1., 1., 1., 1., 1.]])
```

```
In [59]: #full
np.full((4,5), '🐼')
```

```
Out[59]: array([[ '🐼', '🐼', '🐼', '🐼', '🐼'],
                [ '🐼', '🐼', '🐼', '🐼', '🐼'],
                [ '🐼', '🐼', '🐼', '🐼', '🐼'],
                [ '🐼', '🐼', '🐼', '🐼', '🐼']], dtype='<U1')
```